English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the original handwritten solution notes. The original calculations and arguments are retained. Apparent inconsistencies in the original are marked with numbered transcription notes at the end.

Problem 1

1. Conditional probability

A yellow ball means rolling one die, green means two dice, and blue means three dice. Thus

$$\mathbb{P}(\text{green} \mid \text{sum } 3) = \frac{\frac{2}{36}\cdot\frac{1}{3}}{\frac{1}{6}\cdot\frac{1}{6}+\frac{2}{36}\cdot\frac{1}{3}+\frac{1}{216}\cdot\frac{1}{2}} = \frac{8}{21} \approx 38.10\%.$$

2. Five dice

$$\mathbb{P}(\max \geq 3) = 1 - \mathbb{P}(\max = 1 \text{ or } 2) = 1 - \left(\frac{1}{3}\right)^5.$$

For the probability that the product is divisible by 10, let A be the event that at least one number is divisible by 5 and B the event that at least one number is divisible by 2. Then

$$\mathbb{P}(A^c) = \left(\frac{5}{6}\right)^5, \qquad \mathbb{P}(B^c) = \left(\frac{1}{2}\right)^5, \qquad \mathbb{P}(A^c \cap B^c) = \left(\frac{2}{6}\right)^5.$$

$$\begin{aligned}\mathbb{P}(A \cap B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B)\\ &= 1 - \mathbb{P}(A^c) + 1 - \mathbb{P}(B^c) - (1 - \mathbb{P}(A^c \cap B^c))\\ &= 1 - \left(\frac{5}{6}\right)^5 - \left(\frac{1}{2}\right)^5 + \left(\frac{1}{6}\right)^5 \approx 56.70\%. \qquad \text{(Note 1)}\end{aligned}$$

3. Seven balls in seven boxes

For $X_3 = 2$, two boxes contain three balls each and one box contains one ball:

$$\mathbb{P}(X_3 = 2) = \frac{7\binom{7}{3}\,6\binom{4}{3}\,5/2}{7^7} \approx 1.78\%.$$

For $X_3 = 1$, the possible nonzero occupancies are $(3,4)$, $(3,1,1,1,1)$, $(3,2,2)$ and $(3,2,1,1)$. Hence

$$\begin{aligned}\mathbb{P}(X_3 = 1) = {} & \frac{7\binom{7}{3}\,6}{7^7} + \frac{7\binom{7}{3}\,6\cdot5\cdot4\cdot3}{7^7}\\ & + \frac{7\binom{7}{3}\,6\binom{4}{2}\,5/2}{7^7} + \frac{7\binom{7}{3}\,6\binom{4}{2}\,5\cdot4}{7^7} \approx 34.99\%.\end{aligned}$$

The original then writes

$$\mathbb{P}(X_3 = 0) = 1 - \mathbb{P}(X_3 = 0 \text{ or } 1) \approx 63.23\%. \qquad \text{(Note 2)}$$

Check. Enumerate the arrangements with no box containing three balls. The following table transcribes the separate counts and their approximate contributions from the handwritten check. Every numerator is divided by 7^7.

Nonzero occupancies	Numerator	Approximation
$1+1+1+1+1+1+1$	$7!$	0.61%
$2+1+1+1+1+1$	$7\binom{7}{2}\,6\cdot5\cdot4\cdot3\cdot2$	12.85%
$4+1+1+1$	$7\binom{7}{4}\,6\cdot5\cdot4$	3.57%
$2+2+1+1+1$	$7\binom{7}{2}\,6\binom{5}{2}\,5\cdot4\cdot3/2$	32.13%
$5+1+1$	$7\binom{7}{5}\,6\cdot5$	0.33%
$4+2+1$	$7\binom{7}{4}\,6\binom{3}{2}\,5$	2.68%
$2+2+2+1$	$\binom{7}{3}\frac{7\cdot6\cdot5\cdot4\cdot3\cdot2}{(2!)^3}\,4$	10.71%
$6+1$	$7\binom{7}{6}\,6$	0%
$5+2$	$7\binom{7}{5}\,6$	0%
Total		63.23%

$$\mathbb{E}X_3 = \mathbb{P}(X_3=1) + 2\mathbb{P}(X_3=2) \approx \frac{35}{100} + \frac{4}{100} \approx \frac{37}{100}. \qquad \text{(Note 3)}$$

Also, in the notation of the original,

$$\sum_{i=0}^{7} i\mathbb{E}X_i = \mathbb{E}[X_0 + 1X_1 + \cdots + 7X_7] = 7, \qquad \text{(Note 4)}$$

because the weighted occupancy counts give the total number of balls. Similarly,

$$\sum_{i=0}^{7} \mathbb{E}X_i = \mathbb{E}[X_0 + \cdots + X_7] = \mathbb{E}[\text{number of boxes}] = 7.$$

4. Products and quotients

For independent random variables, $\mathbb{E}[XY] = \mathbb{E}X\,\mathbb{E}Y$. See the lectures for the proof. In general, $\mathbb{E}[X/Y] \neq \mathbb{E}X/\mathbb{E}Y$. A counterexample can be constructed from simple random variables, for instance independent X, Y with

$$\mathbb{P}(X=1) = \mathbb{P}(X=2) = \tfrac{1}{2}, \qquad \mathbb{P}(Y=2) = \mathbb{P}(Y=3) = \tfrac{1}{2}.$$

Problem 2

1. A randomly labelled die

Let X be the number of faces labelled 6. Then $X \sim \text{Bin}(6, 1/6)$. Let Y be the outcome of the first roll. By total probability,

$$\begin{aligned}\mathbb{P}(Y=6) &= \sum_{k=0}^{6} \mathbb{P}(Y=6 \mid X=k)\mathbb{P}(X=k) \\ &= \sum_{k=0}^{6} \frac{k}{6}\binom{6}{k}\left(\frac{1}{6}\right)^k\left(\frac{5}{6}\right)^{6-k} = \frac{1}{6}\mathbb{E}[\text{Bin}(6,1/6)] = \frac{1}{6}.\end{aligned}$$

Let Z be the number of rolls until the first 6. A die with no face labelled 6 is possible. Conditional on $X=0$, we have $Z=\infty$: a 6 is never rolled. Consequently, $\mathbb{E}Z = \infty$.

2. A product divisible by 6

The expectation is again infinite, because all the faces might be labelled 5.

3. Two observed sixes

Bayes' formula gives

$$\begin{aligned}
&\mathbb{P}(\text{all faces are 6} \mid \text{two observed 6s}) \\
&\quad = \frac{1 \cdot 1 \cdot \mathbb{P}(X=6)}{\sum_{k=0}^{6} \mathbb{P}(X=k)\mathbb{P}(\text{two observed 6s} \mid X=k)} \\
&\quad = \frac{(1/6)^6}{\sum_{k=0}^{6} \binom{6}{k}(1/6)^k(5/6)^{6-k}(k/6)^2} \\
&\quad = \frac{(1/6)^6}{(1/6)^2\mathbb{E}[\mathrm{Bin}(6,1/6)^2]} \\
&\quad = \frac{(1/6)^4}{6\cdot(1/6)\cdot(5/6)+(6/6)^2} \approx 0.04\%.
\end{aligned}$$

For the most likely number of sixes, compare

$$f(k) = \mathbb{P}(X=k)(k/6)^2 \;\propto\; \binom{6}{k}5^{6-k}(k/6)^2.$$

The respective rescaled values are

k	0	1	2	3	4	5	6
rescaled $f(k)$	0	520.83	1041.67	625	166.67	20.83	1

Thus, two faces labelled 6 is the most likely case.

Problem 3

1. The probability of selecting k

Write $p_n(k) = \mathbb{P}(k \in E_n)$, the probability of selecting k when the first choice is from $\{1,\dots,n\}$. The key observation is

$$p_n(k) = p_n(k+1) + p_n(k+1)\frac{1}{k}.$$

The first term corresponds to selecting k without selecting $k+1$ (why?); the second corresponds to selecting $k+1$ and then k. Hence

$$p_n(k) = \frac{k+1}{k}p_n(k+1).$$

Using $p_n(n) = 1/n$ (or $p_n(1) = 1$), telescoping gives

$$p_n(k) = \frac{1}{k}.$$

2. Conditional probability

$$\mathbb{P}(2 \in E_n \mid 3 \notin E_n) = \frac{p_n(3)}{1-p_n(3)} = \frac{1/3}{2/3} = 50\%.$$

3. Mean and variance

Let $X_i = \mathbf{1}_{\{i \in E_n\}}$. Then

$$\mathbb{E}|E_n| = \mathbb{E}\sum_{i=1}^{n} X_i = \sum_{i=1}^{n}\mathbb{P}(i \in E_n) = 1 + \frac{1}{2} + \cdots + \frac{1}{n} \sim \ln n.$$

For the variance, use

$$D\left(\sum_{i=1}^{n} X_i\right) = \sum_{i,j=1}^{n}\mathrm{Cov}(X_i, X_j), \qquad DX_i = \frac{1}{i} - \frac{1}{i^2}.$$

For $i < j$,

$$\mathrm{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}X_i \mathbb{E}X_j = \mathbb{P}(i, j \in E_n) - \frac{1}{i}\frac{1}{j}.$$

However,

$$\mathbb{P}(i, j \in E_n) = \mathbb{P}(i \in E_n \mid j \in E_n)\mathbb{P}(j \in E_n) = \frac{1}{i}\frac{1}{j} \qquad \text{(why?)}.$$

Thus the covariances vanish for $i \neq j$, and

$$D|E_n| = \sum_{i=1}^{n} DX_i = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \left(1 + \frac{1}{2^2} + \cdots + \frac{1}{n^2}\right) \sim \ln n,$$

since the sum of $1/i^2$ converges.

Problem 4

1. Joint distribution and correlation

The joint distribution is

$Z \backslash X$	0	1	2	$\mathbb{P}(Z = z)$
0	1/9	0	2/9	1/3
1	2/9	1/9	0	1/3
2	0	2/9	1/9	1/3
$\mathbb{P}(X = x)$	1/3	1/3	1/3	1

Therefore $\mathbb{E}X = \mathbb{E}Z = 1$ and

$$DX = DZ = \tfrac{1}{3} \cdot 1^2 + \tfrac{1}{3} \cdot 2^2 - 1 = \tfrac{2}{3}.$$

$$\mathrm{Cor}(X, Z) = \frac{\mathbb{E}[XZ] - \mathbb{E}X\mathbb{E}Z}{\sqrt{DX\, DZ}} = \frac{1 \cdot 1 \cdot \frac{1}{9} + 2 \cdot 1 \cdot \frac{2}{9} + 2^2 \cdot \frac{1}{9} - 1^2}{\sqrt{4/9}} = 0.$$

2. Generating function of a die

$$g_X(s) = \tfrac{1}{6}(s + \cdots + s^6), \qquad \mathbb{E}X = g_X'(1) = 1, \qquad \text{(Note 5)}$$
$$DX = g_X''(1) + g_X'(1) - [g_X'(1)]^2.$$

3. Generating functions and independence

If X and Y are independent, $\mathbb{P}(X = k, Y = m) = \mathbb{P}(X = k)\mathbb{P}(Y = m)$, so

$$\begin{aligned}
\mathbb{E}s^{X+Y} &= \sum_{k=0}^{\infty} s^k \mathbb{P}(X + Y = k) \\
&= \sum_{k=0}^{\infty} s^k \sum_{m=0}^{k} \mathbb{P}(X = m, Y = k - m) \\
&= \sum_{k=0}^{\infty} s^k \sum_{m=0}^{k} \mathbb{P}(X = m)\mathbb{P}(Y = k - m) \\
&= \sum_{k=0}^{\infty} \sum_{m=0}^{k} s^m \mathbb{P}(X = m) s^{k-m} \mathbb{P}(Y = k - m) \\
&= \sum_{m=0}^{\infty} s^m \mathbb{P}(X = m) \sum_{k=m}^{\infty} s^{k-m} \mathbb{P}(Y = k - m) \\
&= g_X(s) g_Y(s).
\end{aligned}$$

The order of summation is changed over the region $0 \leq m \leq k$.

The converse is false! The distribution in part 1 is a counterexample. The original expands the product as

$$g_X(s)g_Z(s) = (1+s+s^2)^2(1/3)^2 = \frac{s^4+2s^3+3s^2+2s+s}{9}. \qquad \text{(Note 6)}$$

Meanwhile,

$X+Z$	0	1	2	3	4
probability	1/9	2/9	3/9	2/9	1/9

Thus $g_{X+Z}(s) = g_X(s)g_Z(s)$, although X, Z are dependent:

$$\mathbb{P}(X=0, Z=2) = 0 \neq \mathbb{P}(X=0)\mathbb{P}(Z=2).$$

4. Alternative dice

Yes, it is possible. For a standard die,

$$g_X(s) = \tfrac{1}{6}(s+\cdots+s^6), \qquad g_{X+Y}(s) = \tfrac{1}{36}(s+\cdots+s^6)^2.$$

We want to express this as the product of two generating functions other than the standard one. Equivalently, factor

$$(s+\cdots+s^6)^2 = P(s)Q(s), \qquad P(1) = Q(1) = 6.$$

The initial list of conditions in the original has $P(0) = Q(0) = 1$ (why?), whereas the following page uses $P(0) = Q(0) = 0$. See Note 7.

$$\begin{aligned}(s+\cdots+s^6)^2 &= [s(s^2+s+1)(s^3+1)]^2 \\ &= [s(s+1)(s^2+s+1)(s^2-s+1)]^2.\end{aligned}$$

At $s=1$ these four factors have values $1, 2, 3, 1$, respectively. The conditions $P(0) = Q(0) = 0$ and $P(1) = Q(1) = 6$ require the factors $s(s+1)(s^2+s+1)$ in both polynomials.

Case 1.

$$P(s) = s(s+1)(s^2+s+1), \qquad Q(s) = P(s)(s^2-s+1)^2.$$

Expanding,

$$P(s) = s^4+2s^3+2s^2+s, \qquad Q(s) = s^8+s^6+s^5+s^4+s^3+s.$$

The dice therefore have faces

$$(1,2,2,3,3,4) \qquad \text{and} \qquad (1,3,4,5,6,8).$$

Case 2.

$$P(s) = Q(s) = s(s+1)(s^2+s+1)(s^2-s+1),$$

which gives two standard dice.

Transcription notes

These notes identify apparent slips in the Bulgarian handwritten source; they are not part of its original solution text.

1. The last term on the final line is written $(1/6)^5$ in the source, but the preceding line gives $(2/6)^5$. The displayed approximation 56.70% agrees with $(2/6)^5$.
2. The source writes "0 or 1"; the complement of $X_3 = 0$ is $X_3 = 1$ or $X_3 = 2$.
3. The source's approximate arithmetic $35/100 + 4/100 \approx 37/100$ is inconsistent. The exact displayed probabilities give $\mathbb{E}X_3 = \mathbb{P}(X_3=1) + 2\mathbb{P}(X_3=2)$.

4. The source includes X_0 in the weighted sum. Its coefficient should be zero when counting balls. The unweighted sum, which counts boxes, does include X_0.

5. The source writes $g'_X(1) = 1$. For the generating function on that same line, the derivative at 1 is $7/2$.

6. The last term of the expanded numerator is written s in the source. Expansion of $(1 + s + s^2)^2$ gives constant term 1, in agreement with the subsequent distribution table.

7. The source initially writes $P(0) = Q(0) = 1$, then uses $P(0) = Q(0) = 0$. The latter is consistent with positive integer die labels and with the factors displayed.